The Psychrometric Chart

HVAC's master diagram — and the missing tool in most Building Management Systems

Why equipment designers cannot work without it, why traditional BMS never adopted it, and how this one chart answers ASHRAE 55, 62.1, 90.1 and Guideline 36 at the same time.

Moist-air states · Design tool · ASHRAE-aligned · Live controls

START HERE

Who uses this chart — and on what equipment

Long before a building is controlled, someone has to choose the machines that condition its air. Those people live on the psychrometric chart.

Who they are

"HVAC equipment designers" are the mechanical engineers who select and size air-conditioning equipment:

- manufacturers' application engineers
- consulting / design engineers
- commissioning engineers who prove it works

What "HVAC equipment" means

The machines that actually change the air:

- air-handling units (AHUs) and their heating / cooling coils
- mixing & economizer dampers
- humidifiers and dehumidifiers
- energy-recovery wheels (ERV / HRV)
- chillers and DX systems, fans, VAV terminals

What they must decide

- How much cooling or heating is needed
- How much of that cooling removes **moisture** rather than heat
- How big the coil must be
- How much outside air to mix in
- Whether condensation or coil freezing will occur

Every one of those decisions is a point or a line on the psychrometric chart. That is why the chart — not the equipment catalogue — is where air-conditioning design actually happens.

WHY IT EXISTS

Why designers cannot work without it

Air conditioning is not simply "making air colder". It is moving **moist air** from one state to another — and the psychrometric chart is the map of every state air can be in.

Every process is a line

Cooling, heating, humidifying, dehumidifying and mixing each move the air in a characteristic direction. Draw the line and you have described the process.

Capacity comes from the chart

The energy content (enthalpy) at each end of the line, multiplied by airflow, is the kW the coil must deliver. That is how equipment gets sized.

Sensible vs latent

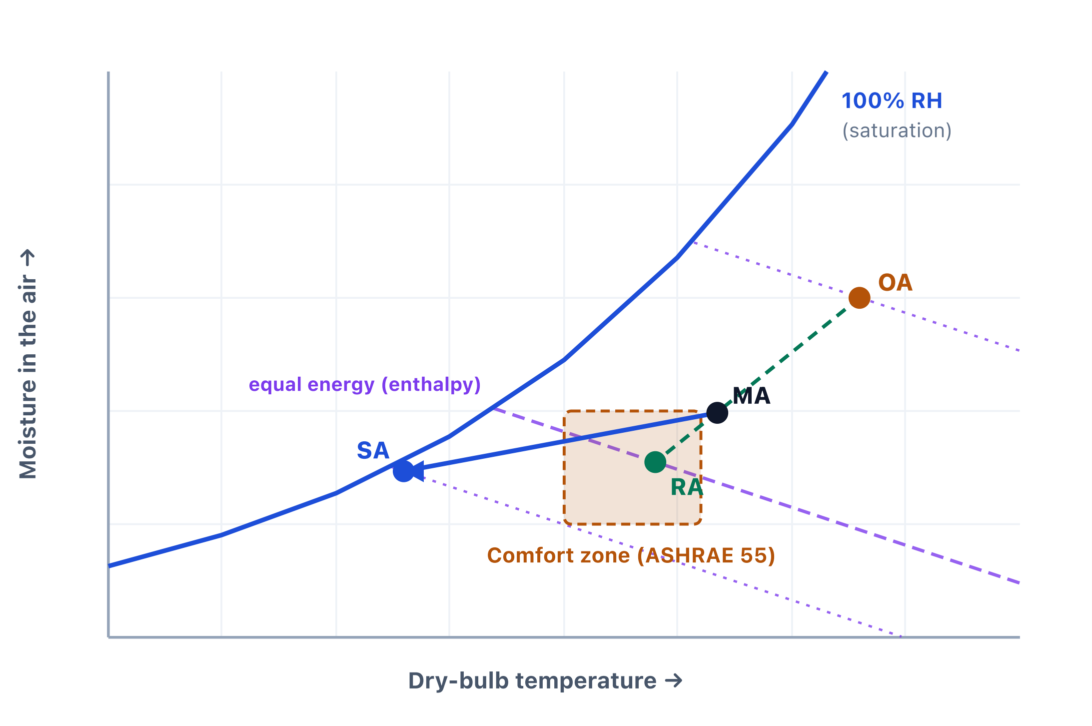
The chart separates cooling that lowers temperature from cooling that removes water. The two behave differently and must be sized separately.

Physical limits are visible

The curved boundary is 100% humidity — air cannot go beyond it. It shows where condensation begins and where a coil would freeze.

HOW TO READ IT

One dot is the state of the air



OA outside air · **RA** return air · **MA** the mixture entering the coil · **SA** supply air delivered to the rooms. Dashed green = mixing; solid blue = the cooling coil. The **violet diagonals** are lines of equal total energy: outside air here holds 79 kJ/kg against return air's 48, so it sits well above RA's line and costs more to condition — no free cooling today.

1 Two axes fix everything

Across is temperature; up is the actual weight of water the air carries. Pin both and the air's state is pinned — relative humidity, dew point, wet-bulb and energy content all follow from that single dot.

2 The curve is a hard limit

It marks 100% humidity — air can never sit above it. Drive the dot onto the curve and water comes out: that is the dew point, a wet coil, a fogged window.

3 Mixing is a straight line

Outside and return air blend to a point that must land on the line joining them. How far along it sits gives the proportion: a third of the way from RA means a third outside air.

4 A coil pulls down and left

Left removes temperature (sensible heat); down removes water (latent heat). The slope of MA→SA is the split between them — and that split is what sizing a coil actually means.

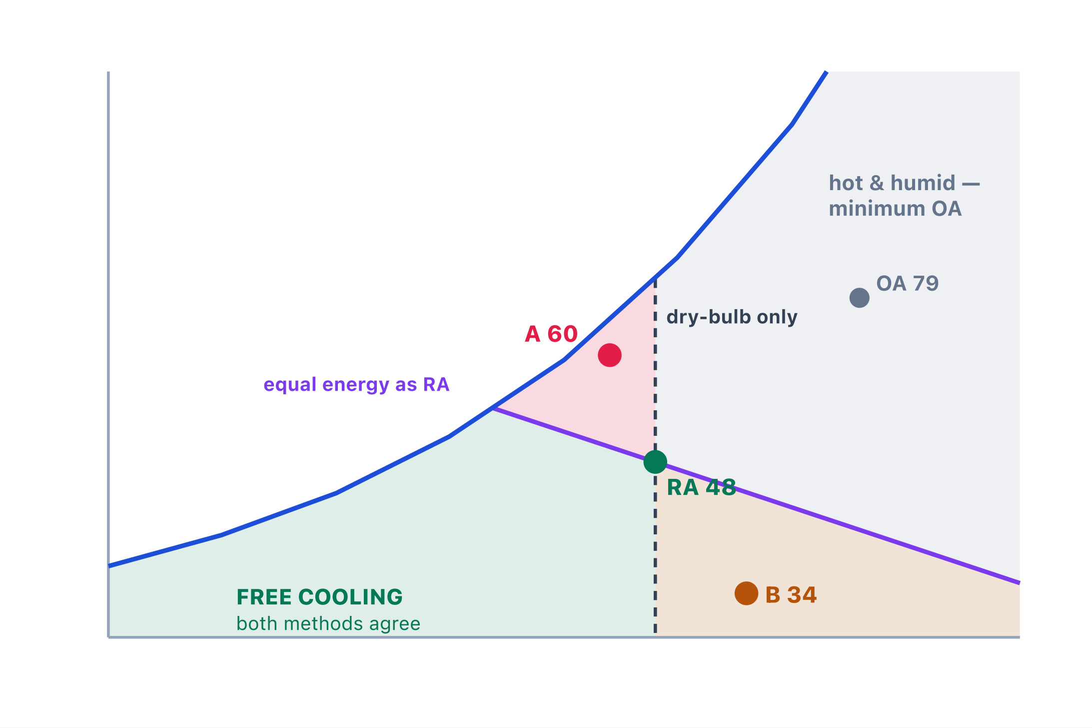
5 Comfort is an area

The acceptable range of temperature and humidity is drawn straight onto the chart. Supply air has to be chosen so the room lands inside that box — not merely at the right temperature.

6 Faults show up as bad geometry

If the mixed-air dot does not sit on the line between outside and return air, something is lying — a stuck damper or a drifting sensor. The picture catches what alarm limits miss.

When is outside air free?



Numbers are total energy in kJ/kg. **RA** 24 °C / 50% RH = 48 · **A** 22 °C / 90% RH = 60 · **B** 28 °C / 10% RH = 34 · **OA** 33 °C / 18 g/kg = 79 (the outside air from the previous slide).

Two lines, four outcomes

1. **Below-left of the violet line** — outside air holds less energy than the air coming back. Open the economizer and let mechanical cooling back off.

2. **Above-right** — outside air is a load, not a resource. Close to the minimum ventilation rate 62.1 demands and let the coil do the work.

A. **Cool but muggy.** At 22 °C it is 2° cooler than the return, so a dry-bulb economizer opens — yet it carries 60 kJ/kg against 48. You have just imported latent load for the coil to wring back out.

B. **Warm but dry.** At 28 °C dry-bulb keeps it shut, yet 34 kJ/kg is far below the return. Energy says use it — though a dry-bulb high limit still guards this corner when the load is mostly sensible.

=. **On the line** the two streams are energy-neutral, so the choice falls to humidity and fan power. Sequences add a deadband here so the dampers do not hunt.

A dry-bulb sensor can only draw the **vertical** line; the chart draws the **diagonal**. The two shaded wedges are precisely where they disagree — and where energy is quietly lost.

THE GAP

Why traditional BMS never adopted the chart

Controls think in single numbers

A BMS is assembled from loops that each watch **one** value — a temperature, a pressure. A state point needs two readings judged together; classic controls had no concept of a two-dimensional air state.

The sensors often aren't there

You need temperature **and** humidity at the same place. Many units measure only dry-bulb on outside, mixed and return air — extra humidity sensors mean extra cost, wiring and calibration drift.

The maths was too expensive

Moisture content and energy content are non-linear and depend on barometric pressure. Older controllers had very little memory and no floating-point maths, so psychrometrics were simply left out.

Graphics couldn't draw it

Early front-ends drew simple equipment schematics and time-series trends. A live chart with curves and overlays needs real rendering — so the chart stayed a static page in the design binder.

Design and operations are separate worlds

The chart is used once, offline, to select equipment. What is handed to the controls contractor is a list of setpoints — not the physics behind them. The reasoning is lost at handover.

The simpler option was allowed

Codes permit dry-bulb-only economizer switching, so designs avoided humidity sensing that was historically unreliable. The easier, compliant path quietly became the default.

The consequence

Operators end up diagnosing a **two-dimensional** problem — heat *and* moisture — by staring at separate one-dimensional trend lines. The information is there; the picture is not.

THE RULEBOOKS

Four ASHRAE standards, in one line each

ASHRAE 55	**Comfort**	Will people feel comfortable?
ASHRAE 62.1	**Fresh air**	Is there enough outside air to stay healthy?
ASHRAE 90.1	**Energy**	Are we doing it without wasting energy?
Guideline 36	**The referee**	How should the equipment be run to hit all three?

Comfort and fresh air push toward **more** conditioning; energy pushes toward **less**. Guideline 36 is the rulebook that settles the argument.

THE BIG IDEA

One sheet — every rulebook draws on it

ASHRAE 55 draws an AREA

The comfort zone: the band of temperature and humidity occupants accept. The whole air-side of the standard becomes a region you must land the room inside.

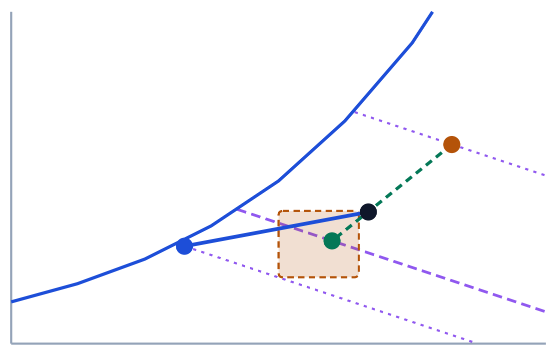
ASHRAE 90.1 draws a DIRECTION

The violet diagonals are equal-energy lines. Which side of them outside air falls on decides whether it is the cheaper source, and how far the dot may be pushed before energy is wasted.

Air-side only — fan power and chiller efficiency sit outside the chart.

The psychrometric chart

temperature × moisture — the only two axes



Every requirement around it becomes a shape on **these same axes**.

ASHRAE 62.1 draws a POINT ON A LINE

The required outside-air fraction fixes exactly where the mixed-air dot must sit along the line between return and outside air.

Guideline 36 draws the PATH

The sequences are the route the dot travels through the day — economizer, coil, setpoint reset — from outside air all the way to the room.

Read this the right way round

The chart is not the small overlap left over where four standards happen to agree. It is the **common sheet** they are all drawn on — an area, a line, a direction and a path, sharing one pair of axes. Satisfy all four and the shapes simply fit together.

HOW IT SATISFIES EACH

One dot on the chart answers every question

55 — Comfort

The comfort zone is drawn straight onto the chart. Is the room dot inside the acceptable temperature/humidity band — and if not, which way is it off?

62.1 — Fresh air

The mixed-air dot must sit on the line between return and outside air. Its position along that line *is* the outside-air proportion.

90.1 — Energy

Equal-energy lines reveal free cooling: when outside air holds less energy than return air, the economizer can meet the load with no mechanical cooling.

G36 — Sequences

Visible proof the sequences land the dot inside the comfort zone, on the correct fresh-air line, along the lowest-energy path.

In one sentence: plot where the air IS, see where it MUST be, how it gets there, and whether it does so at least energy.

THE DIFFERENCE

Trend-based BMS vs a psychrometric-aware BMS

QUESTION BEING ASKED	TREND-BASED BMS (TODAY)	PSYCHROMETRIC-AWARE BMS
What the operator sees	Several separate trend lines, read one at a time	One picture showing the air's actual state
Is the outside-air mix right?	Inferred from the damper command	The mixed-air dot must fall on the outside–return line
Can we free-cool right now?	Usually temperature-only, so humid-but-cool and dry-but-warm hours are misjudged	Decided on true energy content of the air
Humidity control	A separate loop, often fighting the cooling loop	Heat and moisture handled as one state
Finding faults	Alarm limits on individual points	The shape of the process line exposes coil, damper and sensor faults
Proving the standards	Checked manually, usually after the fact	Comfort, ventilation and energy judged together, live

None of this needs new field equipment beyond humidity sensing on the air streams. The difference is what the software does with readings the building is already taking.

PUBLISHED EVIDENCE

What is it worth — and whose credit is it?

What good air-side control is worth

The prize, and it belongs to the sequences. The chart does not produce these savings — it makes them verifiable.

31%

average HVAC saving from Guideline 36 sequences, against existing practice

Berkeley Lab, Spawn of EnergyPlus, medium office, 2022

> Mostly static-pressure reset, zone minimum airflow and scheduling. Only the economizer and SAT logic land on the chart.

41% HEATING **20%** COOLING **18%** SHOULDER

seasonal saving from supervisory setpoint resets

PNNL large-office emulator, Chicago, 2024

> Airside and plant-side combined — the chilled- and hot-water resets sit off the chart entirely.

What the chart itself contributes

Visibility: the missed hours and the broken parts that a temperature trend hides.

9%

median whole-building saving from fault detection, range 1–28%

Berkeley Lab Smart Energy Analytics Campaign, 6,500 buildings

> The standard AHU rule set (NIST **APAR**, 28 mass- and energy-balance tests) is this chart's geometry in algebra. Covers all FDD, so read it as an upper bound.

64% NOT A SAVING

of economizers found faulty in the field

New Buildings Institute; wider surveys average 58%

> Opportunity size, not a return: how often the free-cooling asset is already broken. The chart is how you notice.

Where the saving actually comes from

- Free-cooling hours **taken** rather than missed, because the decision is made on energy rather than temperature alone.
- Outside air delivered at the rate 62.1 requires — not quietly at double it.
- Supply air reset to follow the load instead of one fixed setpoint all year.
- Coil, damper and sensor faults caught in days rather than at the next seasonal complaint.

Read these numbers honestly

- Measure-level results from simulation and field studies — not a guarantee for any one building.
- They overlap and are not additive; the left pair and the right pair count much of the same energy twice.
- The largest savings come from the worst starting points.
- Enthalpy switching is **not** automatically better — Taylor & Cheng (2010) found fixed dry-bulb often wins once humidity sensors drift, hence 90.1-2013's accuracy limits.

The chart itself saves nothing. It makes the opportunity visible and the fault obvious — the saving is booked by the sequence that acts on it.

One chart. Four rulebooks.

The psychrometric chart is not a design-only artifact. Bringing it into the building's controls turns anonymous setpoints back into visible physics — and puts comfort, air quality and energy into a single picture an operator can read at a glance.

The computing power to draw it live has been ordinary for years. What has kept the chart out of the control room is habit and missing humidity sensors — not physics, and no longer technology.